

Impact of Macroeconomic Variables on Infant Mortality in Nigeria (1990-2024)

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Article DOI: 10.48028/iiprds/ijsressd.v6.i1.07

Abstract

Good health is indispensable and improves the productivity of human capital which is a game changer and a key factor in achieving sustainable economic growth and development. This paper examines the impact of Macroeconomic variables particularly shedding light on access to electricity, on Infant Mortality Rate in Nigeria. This paper explores the direct and indirect relationship between macroeconomic variables namely: Gross Domestic Product Per Capita (GDPPC), Government Health Expenditure (GHE), Access to Electricity (ACE) and Health-Specific Aid and health indicator such as infant mortality rate. The Autoregressive Distributed Lag (ARDL) technique was used to examine the short- and long-run impact of macroeconomic variables on infant mortality rate. The health outcome was captured by infant mortality rate, utilizing econometric analysis of panel data from 1990 to 2024. The findings reveal that GDPPC, GHE, and AID showed the mix of statistically significant and insignificant impact on the infant mortality while access to electricity (ACE) was significant. This implied that they improve infant mortality by enhancing the effectiveness of healthcare services, improving living standards, and fostering economic growth. The paper laid emphasis on the importance of not only expanding electricity coverage but also improving the quality of supply to achieve better health outcomes. Policy recommendations are provided to guide targeted investments in energy infrastructure and macroeconomic stability as crucial components of health improvement strategies in Nigeria.

Keywords: *Macroeconomic Variables, Infant Mortality, Access to Electricity, Nigeria.*

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<https://internationalpolicybrief.org/international-journal-of-scientific-research-in-educational-studies-social-development-volume-6-number-1/>

Background to the Study

The concept of sustainable development stresses that infant mortality rates are directly impacted by a successful strategy that integrates economic growth, social equity, and environmental protection. United Nations (2015), on that note prioritized good health and well-being which focused mainly on universal health coverage, reduced infant mortality, and combating communicable diseases as captured in their Sustainable Development Goals (SDGs) as in the 2030 Agenda for Sustainable Development. This is streamlined to Brundtland Commission, initially referred to as the World Commission on Environment and Development (WCED). The commission was chaired by Gro Harlem Brundtland, the former Prime Minister of Norway, and it released its influential report titled “Our Common Future” in 1987. It is expected that by 2030 these goals will be made a reality owing to substantial commitment by all stakeholders. Meanwhile in World Bank (2025), an open data on infant mortality rate vividly shows that so far following the 2030 agenda, Norway compared to Nigeria has one of the lowest infant mortality rates due to its strict adherence and commitment to achieving these goals with the infant mortality rate of 2 to 3 deaths per 1000 live births compared to Nigeria's high infant mortality rates of 55 to 75 deaths per 1000 live births. This is due to sophisticated healthcare and social welfare system on the side of Norway whereas Nigeria is plagued by issues related to poor healthcare infrastructure, poverty, and corruption.

The role of technology in achieving standardized health facilities and improvements in the Healthcare service and health outcomes such as infant mortality cannot be overlooked. In Solow (1956), the growth model gives a comprehensive groundwork on how economic growth driven by capital accumulation, labor, and technological progress influences sustainable development including health. Fundamentally, Solow model, highlights technological progress as the primary driver of sustained economic growth in the long run. Nigeria has paid its dues since the millennium declaration of 8th September, 2000 and has committed to various health policies coupled with macroeconomic reinforcing interventions but despite the effort put out, the rate of infant mortality remains significantly high when compared to the international standards. The macroeconomic indicators such as healthcare expenditure, GDP per capita, inflation, and poverty levels, are pivotal in the improvements of health outcomes, most especially the infant mortality rates (IMR). However, increased healthcare spending and targeted health programs have brought about the reduction in IMR from 112 deaths per 1,000 live births in 2000 to an impressive 55 to 60 deaths per 1000 live births in 2023 (Macrotrends, 2024). The progress has been marred by insufficient funding and inefficiencies, poor healthcare infrastructure, and economic instability. It is evident that there's ever-deepening gap between health policy goals and IMR. Thus, there's need for a clarion call to critically examine how macroeconomic health indicators impact infant mortality trends. Therefore, on a broader perspective, this paper looks to empirically examine the impact of macroeconomic Variables on infant mortality rate in Nigeria whereas the specific objectives are to:

- i. Assess the extent of the impact of GDP per capita on infant mortality in Nigeria.
- ii. Examine the degree to which government health expenditure have impacted infant mortality.
- iii. Ascertain the impact of access to electricity on infant Mortality.
- iv. Figure out the impact of health-specific aid on infant mortality.

Materials and Methods

Conceptual Review

Macroeconomic variables are those economic factors and indicators that have a reciprocating relationship with the health sector. This is to say that these variables influence the health sector and in turn is influenced by the health sector. These indicators are to be considered in the provision of healthcare service and the procurement of healthcare products that may determine the status of national health (Naik, *et al.* 2017). According to the stats disclosed by Nigerian Bureau of Statistics (2019), it is shown that GDP Per Capita is measured by the division of GDP by the total population of the Nation. Higher GDP per capita generally shows and or determines the standard of life all over the globe, which represents the capacity of the populace acquiring more goods and services. Therefore, when a country has a high GDP per capita, it implies that the citizens of that country can acquire better healthcare services, housing, quality education, and food. Having access to quality education means better job opportunities and higher levels of employment and income and a high probability of reduced infant mortality rate. Healthcare expenditure simply means the sum amount of money value spent on healthcare services and products for a specific period of time, usually per annum. This money spent exclusively comprises the spending on hospital services, physician and clinical services, dental services, and other health-related expenses like pharmaceutical products, nursing homecare, and administrative costs. World Health Organization (WHO) reports on global health expenditure, global health expenditure has been on a rapid rise, jeopardizing the progress of the plans set to attain a sustainable development and quality access to health care services (WHO, 2021). According to Mohammed (2024) in Nigeria where over 80% of individuals pay out-of-pocket for healthcare, that the direct increase in essential needs of life such as food potentially means that more Nigerians are at cross-road where they must prioritize daily survival over the purchase of healthcare services and products, mostly in health conditions that are asymptomatic or without immediate repercussions.

Aonat (2024), stressed that the importance of technology in the Healthcare system, noting that technological advancement encompasses all forms of innovation in all fields and sectors such as: information technology, energy, and materials and science, et cetera. Thus, electricity as a technology is essential in the functioning of the healthcare service as it enables efficiency in the use of medical equipment, information and communication technologies, safe delivery, neonatal care, storage of vaccines and access to clean water (Ritchie, 2021). Health aid is the financial support received by a country which assist in the development of its healthcare system. However, this aid for Nigeria is a critical supplement in addition to the government healthcare expenditure. This health aid majorly come from – bilateral and multilateral donors, grants, public-private partnership, NGOs, etc. (Mass life Healthcare Limited, 2025). Health outcomes are health consequences which is the result as perceived by a patient post-interface with the existing healthcare system. However, Infant Mortality is a critical indicator of the overall national health status and the effectiveness of healthcare system. Infant Mortality Rate refers to the rate of Infant deaths (under one year) per 1000 live births. According to World Bank (2023), at about 72 out of every 1000 live born infants don't survive their first year of life in Nigeria. In other words, 72 new born babies do not make it past their first year after delivery.

Empirical Review

In the quest to discover the many pitfalls in the realm of healthcare and to proffer the best possible solutions, numerous researchers in their works show how macroeconomic determinants such as – Gross Domestic Product Per Capita (GDPPC), Capital Healthcare Expenditure (RGHE), Access to Electricity (ACE), and Health-Specific Aid (AID) as they relate to health outcomes especially infant mortality rate. These previous studies have critically laid foundation and hence, this study aims to discover the stones left unturned in this area of study. For instance, Attamah (2018), used Vector error correction model (VECM) a statistical technique to analyze impact of government health expenditure on infant mortality rate for the period 1981-2018. The result disclosed a positive and significant impact of the government expenditure on health on infant mortality. and prescribed a health scheme which should be funded by the three tiers of governments that is federal, state, and local.

Also, Ubi and Ndem (2019), used Vector Autoregressive Econometric (VEC) approach to examine whether health outcomes (life expectancy and infant mortality rate) in Nigeria responds to poverty shock. Empirically the study indicates that poverty shock has no direct effect on health outcomes variable. Rather the study showed that variations in health outcomes are not actually due to poverty shocks but due to the health outcome shocks. The recommendation emphasized the improvement in health outcomes of the population is indispensable for poverty reduction. In the same vein and using a broader variable, Raji (2020), used the fully modified OLS technique to investigate the role of macroeconomic factors on infant and under-five mortality rates in Nigeria for the period 1986–2017. the study revealed that the unfriendly macroeconomic policy focused on indicators like the instability of inflation rate, unemployment and exchange rate contributed to the worsened levels of infant and under-five mortality rate in Nigeria. Alternatively, Boachie *et al.* (2020), in a study observed the importance of health of expenditure by the government in low-income and middle-income countries. fixed effects estimation technique was employed to examine this effect during the period from 1995-2014. The result highlighted the importance of public health expenditure as it is pivotal in the reduction of infant mortality. the result further revealed that as socioeconomic conditions such as access to clean water, income per capita, etc., health outcomes such as infant mortality improves. Even Adeosun and Faboya (2020), focused its empirical analysis on the effects of expenditure in healthcare on child mortality in Nigeria, and the outcome proved that there exists a negative correlation between the healthcare expenditure by government and infant mortality which simply translates to spending on healthcare has no significant impact on infant mortality rate in Nigeria.

Adeleke, et al. (2022), employed spatial statistic as the technique to examine the macroeconomic predictors of under-5 mortality in Nigeria. It was discovered that in the northern region of Nigeria that in some areas exhibited high rates of under-5 mortality which is linked to variations in the internally generated revenue where as in the southern region of Nigeria, the inflation rate is observed to influence the under-5 mortality rate negatively. The study recommends a specially designed intervention which is specific to these regions that addresses the respective issues of internally generated revenue and inflation to solve the problem of under-five mortality in Nigeria. Jiang and Liu (2022), examined the causal

relationship between inflation and infant mortality in China which deployed a mixed frequency vector autoregressive model (MF-VAR) not considering any filtering procedure, along with impulse response analysis and forecast misspecification variance decomposition, and hence, compares it with a low frequency vector autoregressive model (LF-VAR). The result showed that a causal relationship exists between inflation and infant mortality. More notably, it was established that the contribution of Consumer price index to infant mortality rate is greater whereas in the forecast error variance decomposition in the MF-VAR model compared to the LF-VAR model, indicating that Consumer price index has stronger explanatory power for infant mortality rate in mixed-frequency data which in national realities explains why it is important to maintain price stability and key macroeconomic indicator in achieving sustainable development goals (SDGs) and therefore, Policymakers should pay serious attention to inflation as a macroeconomic variable that may potentially impact infant mortality negatively. Akpesiri and Eburajolo (2022), investigated the monetary policy and the health sector with focal point on infant mortality outcome in Nigeria between 1980 and 2016. employing the Vector Error Correction Model, the result showed that literacy rate and Per Capita income exerted a negative effect on the infant mortality rate although they were statistically not significant. the study recommends that government should focus more a monetary policy to simulate investment level in the Nigeria's educational system and to as well enhance sustained economic growth in the economy which would in turn reduce infant mortality rate and bring about economic development.

Koginam and Anthony (2023), carried out a study that focused on establishing the relationship among under-five mortality, infant mortality, adult mortality, thereby revealing the behavior of their relationship in a long-run. Using an Autoregressive Distributed Lag (ARDL) approach, the findings disclosed that under-five mortality, adult male mortality, infant mortality and adult female mortality exhibits a negative but statistically insignificant on RGDP in the long run. However, infant mortality has a negative and significant influence on RGDP in the short run, while the other dimensions of mortality exerted a negative and statistically insignificant effect on RGDP in the short run. Also, Awoyemi et al. (2023), investigated the linkages between government expenditure and health outcomes in Nigeria for a period of 23 years (1995-2018). Autoregressive Distribution Lag technique was deployed to assess the short-run and long-run effects of government health expenditure on health outcomes separately where the health outcome tested was life expectancy at birth and mortality rate. By the end of the study, it was discovered that a negative relationship exists between the health expenditure and mortality rate recommending that government must increase its health expenditure and establish a public-private partnership to improve the health care sector and outcomes.

Udemba et al. (2023), asserted that renewable energy sources, social (GDP per capita) and public subsidies (general government final consumption expenditure) have lessening effect on infant mortality in a study on India. The study adopted the use of Relevant scientific methods residual Augmented Least Squares (RALS), Engle and Granger (EG), and its newly augmented version (RALS-EG) and indeed revealed that in India renewable energy sources, social (GDP per capita) and public subsidies (general government final consumption

expenditure) have negative effect on infant mortality in India, and on the contrary, the private subsidies (gross capital formation), fossil fuels, and carbon dioxide cause an increase in infant mortality. Miftahu and Kawu (2024), assessed the impact of healthcare investment on infant mortality in Nigeria. the study utilized the Autoregressive Distributed Lag Model (ARDL) Bound test technique to ascertain long-run relationship between public healthcare investment and infant mortality as well as examine the impact of healthcare investment relating to child support programs, children immunization, improving Nutrition, and education on infant mortality in Nigeria. this study revealed a long-run relationship between healthcare expenditure and infant mortality in Nigeria for the period of study and recommends that government should mobilize and monitor disbursement of funds meant to finance programs that will support child rights, healthy development, reduced infant mortality, and general wellbeing of the children.

Theoretical Framework

The empirical analysis of this paper will be based on the postulation of the Solow-Swan Growth Theory of 1956. As the pioneer of the neoclassical growth theory, the theory fundamentally states that for an economy to grow, three factors are to be considered such as – labor, capital, and technology. Resources are usually scarce in terms of capital and labor meanwhile, the impact of technology to economic growth is extraordinary (Banton, C. 2023). Furthermore, this growth theory emphasized the importance of accumulation of capital in an economy and how this capital is utilized as it encourages economic growth and hence, determines the performance and productivity of labor. Drago lava, et al. (2016), examined the part technology plays in neoclassical growth theory and ascertained that technological progress is the key generator of economic growth and recommended government intervention via investing in scientific and research development towards innovation. According to Corporate Finance Institute (2024), the standard aggregate Cobb-Douglas function of production used by neoclassical growth theorists to measure growth and equilibrium of an economy is:

$$Y = A \cdot K^{\alpha} \cdot (L)^{1-\alpha} \quad 1$$

Where: Y denotes the GDP, A stands for the determinants(s) of technology, K denotes the capital stock, L stands for labor inputs, and Alpha (α) represents the output elasticity of capital usually standing between zero and one. The study focuses on the impact of health macroeconomic variables on infant mortality. Therefore, shifting the focus on economic growth using infant mortality as the proxy and integrating health as a form of human capital infant mortality can be influenced by capital accumulation for example government health expenditure and building of infrastructures, labor growth for instance the health status of the populace, and Technological advancement such as – steady electricity and medical innovations (Makiew, Romar, Weil. 1992). Hence, the modified model for infant mortality:

$$Y = A \cdot k^{\alpha} \cdot (H \cdot L)^{1-\alpha} \quad 2$$

Where: Y denotes for infant mortality rate, A captures the effect technological advancement, k denotes capital (government health expenditure and infrastructures), H denotes human

capital (population health status), and L denotes labor (population size), and Alpha(α) stands for output elasticity of capital.

Methodology

Sources of Data and Method of Analysis

This paper utilized time-series data during analysis thus, Annual time series data from 1990 to 2022 was collected from the Central Bank of Nigeria (CBN) Statistical Bulletin and World Bank. This paper also employed an ex-post facto design which is perfect in tracking changes overtime, evaluating policy or event impacts, and predictive analysis using past data. Autoregressive Distributed Lag (ARDL) was the choice technique for analyzing the impact of health macroeconomic factors on infant mortality rate (IMR) in Nigeria. This will facilitate the determination of short-run and long-run relationships among the variables and for estimating variables integrated into 1(1) and 1(0) (Pesaran & Pesaran 1997; Pesaran & Smith 1999; Pesaran, *et al.* 2001). To estimate and analyze the long- and short-term effects of government health expenditure on Nigeria's health sector growth, this paper used the autoregressive distributed lag (ARDL) and error correction model (ECM). The co-integration of macroeconomic variables and infant mortality in Nigeria was examined using the autoregressive distributed lag (ARDL)-bounds test. The analytical program for model estimation is E-Views 12.

Model Specification

Based on the theoretical exposition of the 1956 Solow-Swan Growth Theory, the preferred mathematical specification that explains the projections and hypotheses are captured in the following models. The Autoregressive Distributed Lag (ARDL) Bounds test for cointegration between the health macroeconomic variables and infant mortality rate in Nigeria from 1990-2024, the implicit function and model are as follows:

$$IMR = f(MV) \quad 1$$

Where: IMR is Infant Mortality Rate and MV is Macroeconomic Variables. Factoring in the variables for Health Macroeconomic variables, gives:

$$IMR = f(GDPPC, CHE, ACE, AID) \quad 2$$

Where: GDPPC is Gross Domestic Product Per Capita, RGHE is Capital Health Expenditure, ACE is Access to Electricity, and AID is Health-specific Aid

Setting up equations in econometric form explicitly gives:

$$IMR = \beta_0 + \beta_1 GDPPC + \beta_2 CHE + \beta_3 ACE + \beta_4 AID + \mu_1 \text{ ---} \quad 3$$

Where: β_0 stands for the intercept or the constant value or the autonomous variable estimates for the Infant Mortality Rate; $\beta_1 - \beta_4$ are the coefficients of the health macroeconomic variables Gross Domestic Product Per Capita (GDPPC), Capital Health Expenditure (CHE), Access to Electricity (ACE), and AID is Health-specific Aid; and μ_1 is the error term. The

Autoregressive Distributed Lagged (ARDL) model that will be used in this paper the short-run form is specified as follows:

$$\Delta IMR_t = \alpha_0 + \sum_{f=1}^a \alpha_f \Delta IMR_{t-f} + \sum_{h=0}^b \theta_h \Delta GDPPC_{t-h} + \sum_{i=0}^c \tau_i \Delta CHE_{t-i} + \sum_{j=1}^d \vartheta_j \Delta ACE_{t-j} + \sum_{k=0}^e \varphi_k \Delta AID_{t-k} + \gamma EC_{t-1} + \varepsilon_t \quad \dots 4$$

$$IMR_t = \beta_0 + \sum_{h=0}^b \theta_h \Delta GDPPC_{t-h} + \sum_{i=0}^c \tau_i \Delta CHE_{t-i} + \sum_{j=0}^d \vartheta_j \Delta ACE_{t-j} + \sum_{k=0}^e \varphi_k \Delta AID_{t-k} + \mu_t \quad \dots 5$$

Where: Δ is the change in infant mortality rate and the independent variables, α_f , θ_h , τ_i , ϑ_j , and φ_k are the short-run coefficients whereas θ_h , τ_i , ϑ_j , and φ_k , are the long-run coefficients, EC_{t-1} in equation 4 is the error correction term that indicates the speed of adjustment to equilibrium, γ determines the rate with which IMR returns to its long-run equilibrium after short-run fluctuations, ε_t and μ_t stands for error term and disturbance term respectively. Equation 4 will be used to examine the short-run and the speed above is used to adjust the estimation until the ECM turns negative while equation 5 is the long-run form. The negative sign of the coefficient of the error correction term ECM (-1) shows the statistical significance of the equation in terms of its associated t-value and probability value.

Descriptive Analysis and Summary Statistic of the Variables

Table 1: Descriptive Analysis and summary statistic of the Variable

	IMR	GDPPC	CHE	ACE	AID
Mean	91.94497	1636.231	132.4343	48.70229	344.5429
Median	92.25500	1872.000	81.91000	49.97000	430.0000
Maximum	125.8740	3201.000	434.7900	70.00000	604.0000
Minimum	53.70000	494.0000	0.150000	27.30000	50.00000
Std. Dev.	25.40127	812.6397	139.0126	9.075104	221.1146
Skewness	-0.050406	-0.013917	0.782693	-0.095740	-0.213676
Kurtosis	1.546022	1.784116	2.292091	2.755247	1.280926
Jarque-Bera	3.097815	2.157093	4.304375	0.140830	4.576022
Probability	0.212480	0.340089	0.116230	0.932007	0.101468
Sum	3218.074	57268.07	4635.200	1704.580	12059.00
Sum Sq. Dev.	21937.63	22453030	657033.2	2800.156	1662317.
Observations	35	35	35	35	35

Source: Output from e-views 12 (2025)

The descriptive analysis, as reflected by the standard deviation mean, kurtosis, and skewness, is presented in Table1. The average Infant Mortality Rate is 93.06 within the ranges 54.74-125.87 and skewness and kurtosis of -0.08 and 1.56 respectively. GDPPC recorded an average of 1658.6, which ranges between 494 and 3201 with skewness and kurtosis of -0.07 and 1.81, meanwhile CHE has an average of 132.4 within the range of 434.8 and 0.15 and with skewness and kurtosis of 3.87 and 16.57. Furthermore, Access to electricity has an average of 48 with skewness and kurtosis of -0.36 and 2.43 and AID recorded an average of 344.5, which ranges between 50 and 604 with skewness and kurtosis of -0.214 and 1.28.

Stationary Tests (Unit Root Tests)

This section of the paper shows the unit root of the variables using the Augmented Dickey-Fuller (ADF) and Philip Perron Test to check the stationary at a 5 percent level of significance.

Table 2: Unit Root Test Result

variables	Test	ADF Values	Critical Values	Order	Test	PP Values	Critical Values	Order
IMR	ADF	-2.938997*	-2.636901	I (0)	PP	-5.128678*	-2.634731	I (0)
GDPPC	ADF	-4.273570*	-2.636901	I (1)	PP	-4.182759*	-2.636901	I (1)
CHE	ADF	-3.905995**	-3.587527	I (1)	PP	-8.886357*	-3.646342	I (1)
ACE	ADF	-2.790636*	-2.636901	I (0)	PP	-3.040229*	-2.634731	I (0)
AID	ADF	-2.226164**	-1.951332	I (1)	PP	-2.463268**	-1.951332	I (1)

* Implies significant at 1% level, **implies significant at 5% level and *** implies significant at 10%.

Source: Author's Computation Using EViews-12 (2025).

The Augmented Dicky Fuller and Philip Perron unit root tests were conducted, as shown in Table 3. Mortality Rate and is non-stationary at level, meaning they are integrated of order 1 they become stationary at first differencing. The remaining variables are stationary at either level or first difference. Following the outcome of the respective results obtained from ADF and Philip perron as well as the order of integration of the variables, the long-run relationship among the variables will be tested using the ARDL model, which can capture the characteristics of a mixture of I (0) and I (1) of the variables as postulated by Pesaran et al. (2001).

Co-integration of ARDL-Bounds Test

This section shows the ARDL co-integration bounds test of the variables used in this paper.

Table 3: ARDL-Bounds Testing

Null Hypothesis: No long-run relationship exists				
Test Statistic	Value	Significance	I (0)	I (1)
F-statistic	162.7671	10%	2.2	3.09
k	4	5%	2.56	3.49
		2.5%	2.88	3.87
		1%	3.29	4.37

Source: Author's Computation Using EViews-12 (2025)

As observed in the table 4 above shows the ARDL bounds test for co-integration that was carried out for all models based on the research objectives F-statistics value of 162.7671 is greater than the lower and upper bound critical values at 1%, 2.5%, 5%, and 10%. Therefore, the null hypothesis of no co-integrating relationship is rejected. This means a long-run relationship exists between Infant Mortality Rate and Macroeconomic variables.

Estimation Results

This section presents the long-run and short-run results of the ARDL regression analysis, where infant mortality rate (IMR) in Nigeria is the dependent variable while Gross Domestic

Product Per Capita (GDPPC), Recurrent Government Health Expenditure (RGHE), Access to Electricity (ACE) and Health-Specific Aid (AID) are the independent variables measuring the macroeconomic variables in Nigeria.

Table 4: ARDL Regression Results

Dependent Variable: IMR

Co-integrating Estimates (ECM Estimates)

Variable	Coefficient	Std. Error	t-Statistic	Prob.*
IMR (-1)	0.893954	0.044039	20.29891	0.0000
GDPPC	0.000592	0.000142	4.159451	0.0013
GDPPC (-1)	-5.93E-05	0.000170	-0.349357	0.7329
GDPPC (-2)	1.14E-05	0.000155	0.073032	0.9430
GDPPC (-3)	4.93E-05	0.000168	0.293812	0.7739
GDPPC (-4)	-0.000328	0.000133	-2.472568	0.0294
CHE	-0.001133	0.001284	-0.882609	0.3948
CHE (-1)	0.001448	0.001486	0.974765	0.3489
CHE (-2)	-0.001865	0.001669	-1.117510	0.2857
CHE (-3)	-0.001399	0.001690	-0.827523	0.4241
CHE (-4)	0.004789	0.001403	3.413318	0.0051
ACE	-0.040936	0.017438	-2.347477	0.0369
ACE (-1)	-0.077347	0.019766	-3.913185	0.0021
ACE (-2)	-0.046867	0.021770	-2.152840	0.0524
ACE (-3)	-0.014440	0.021490	-0.671929	0.5144
ACE (-4)	-0.037625	0.017690	-2.126892	0.0549
AID	-0.002832	0.000841	-3.365440	0.0056
AID (-1)	-0.003434	0.001667	-2.059901	0.0618
CointEq (-1) *	-0.106046	0.002851	-37.19570	0.0000
R-squared	0.999975			
Adjusted R-squared	0.999937			
Durbin-Watson stat	2.949797			
F-statistic	26327.80			
Prob(F-statistic)	0.000000			

Source: Author's Computation Using EViews-12 (2025)

From Table 4, the value of F-statistics of 26327.80 and the probability values of 0.0000 indicated that there is a long-run relationship between health macroeconomic variables and infant mortality in Nigeria. The one period ECT (-1) value of -0.106 in the table was found to be highly significant at 1%. This implies that when there is a case of disequilibrium in the Infant Mortality Rate model, it would take a speed of about 10% to reach adjustment to equilibrium from short-run to the long-run. As seen in the table, it further revealed that the Adjusted R-squared of 0.999937 explained that the macroeconomic determinants accounted for about 99% change in the Infant Mortality Rate. Meanwhile, the remaining 0.000063% showed the other factors affecting the dependent variable not captured. Thus, showing a good fit of the model.

Table 5.

Long Run				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
GDPPC	0.002508	0.001386	1.808857	0.0956
CHE	0.017357	0.045567	0.380902	0.7099
ACE	-2.048302	0.512111	-3.999721	0.0018
AID	-0.059087	0.013541	-4.363389	0.0009

Source: Author's Computation Using EViews-12 (2025)

GDP per capita showed a positive relationship and appeared to be statistically significant in the long while capital health expenditure showed to be positive and statistically insignificant in the long-run. However, for access to electricity, the result showed a Negative relationship which appeared to be statistically significant. Also, the health-specific aid was found to be statistically significant and showing a negative relationship in the long-run.

Hypotheses Testing

Table 6: Hypotheses Testing of ARDL Results

Hypotheses	Test Statistic	Value	Diff	Probability	Remark
$H_{01}: \beta_1 = 0$					
$H_1: \beta_1 > 0$	F-statistic	184.4389	(4, 12)	0.0000	Rejected
$H_{02}: \beta_2 = 0$					
$H_2: \beta_2 > 0$	F-statistic	4.270960	(4, 12)	0.0223	Rejected
$H_{03}: \beta_3 = 0$					
$H_3: \beta_3 > 0$	F-statistic	4.913896	(4, 12)	0.0140	Rejected
$H_{04}: \beta_4 = 0$					
$H_4: \beta_4 > 0$	F-statistic	15.31302	(1, 12)	0.0021	Rejected

Source: Author's computations using Eviews-12 (2025).

Table 6 shows the hypotheses of the impact of health macroeconomic variables on infant mortality rate in Nigeria. Given the respective statistics values for hypotheses 01-05 and their corresponding p-values which are significant under 5% level of significance. Therefore, the null hypothesis that these health macroeconomic variables have no significant impact on infant mortality in Nigeria are rejected implying that the given health macroeconomic variables have significant impact on infant mortality in Nigeria.

Post-Diagnostic Checks

Table 7: Results of Post-Diagnostic Checks

Test		Outcomes	
		Coefficients	Probability
Breusch-Godfrey	F-stats	1.527989	0.2822
Serial Correlation LM Test			
Heteroskedasticity:	F-stats	0.894580	0.5964
Breusch-Pagan-Godfrey			
Normality Test	Jarque-Bera	1.972475	0.372977

Source: Author's Computation Using EViews-12 (2025).

Table 7 showed that there is no serial correlation since the F-statistics is 0.1.3 and with the P-value of 0.29 that is greater than the 5% significance level. This outcome suggests the absence of serial correlation in the model of the impact of health macroeconomic variables on infant mortality in Nigeria. Similarly, the heteroskedasticity results show that variables are free from the problem of Heteroskedasticity since the F-statistics of 0.89 and P-value of 0.596 are greater than the 5% significance level. This outcome suggests the absence of heteroskedasticity in the model of the impact of health macroeconomic variables on the infant mortality in Nigeria. Also, the Jarque-Bera test of normality shows that the error term in our specified equation is normally distributed. Finally, this is seen as the respective insignificant Jarque-Bera statistics of 1.97 and the probability value of 0.37.

Implication of Findings

The empirical findings showed on the short-run that there's a positive and significant effect of the explanatory variables on infant mortality in Nigeria. On the long-run, it showed the mix of statistically significant and insignificant impact on the infant mortality. Access to Electricity was seen to have a negative effect on Infant mortality outcomes and appeared to be statistically significant which relates to the notion that stable access to electricity may reduce infant deaths given that the necessary policies are made. The high Adjusted R-squared coefficient is a proof that health macroeconomic variables explain the variations in the. These findings align with the work of Awoyemi *et al* (2018) who discovered that there exist a significant impact of technology thereby reducing persistent challenges in the Healthcare system and improving health outcomes.

Conclusion and Recommendations

This paper employed the Autoregressive Distributed Lag (ARDL) model and the Error Correction Model (ECM) to examine the long-term relationship between health macroeconomic variables, particularly access to electricity, and infant mortality in Nigeria. The results confirm the existence of a significant long-term relationship between these variables. Specifically, the findings indicate that improvements in electricity access and macroeconomic stability—measured through indicators such as GDP per capita and inflation—positively impact infant mortality, including reductions in infant mortality and improvements in life expectancy. The error correction term was found to be statistically significant and negative, indicating that deviations from the long-term equilibrium are corrected over time. The study highlights the critical role of energy access as a fundamental driver of health and development, suggesting that targeted investments in this area can yield significant long-term economic benefits. Therefore, the paper recommended the following:

- i. Nigeria should focus on improving the health budget by meeting the recommended 15% of national budget by the African Union since good health is associated with productive capacity.
- ii. Policy makers should focus on improvement of health if growth is to be sustained. This can be achieved through the control of these health macroeconomic indicators by implementing policies that are friendly to the health sectors and to ensure economic growth and development such as to meet the Sustainable Development Goals (SDGs) Agenda for Sustainable Development before the year 2030.

- iii. A stable and sustainable access to electricity by investing in off grid energy source such as renewable energy like solar which plays an important role in avoiding infant deaths in Nigeria. Electricity facility also helps individuals to be health conscious via electricity run devices, media. and it encourages proper medical treatment with modern electricity led medications. For this reason, an effective and efficient policy formulation to ensure more electricity access should be formulated that will be conducive for national health status.

References

- Adeleke, R., Osayomi, T., Iyanda, A. E., & Alabede, O. (2022). Spatial analysis of economic predictors of under-five mortality in Nigeria, *Open Health*, 3(1), 103–113. <https://doi.org/10.1515/openhe-2022-0012>
- Adeosun, O. T. & Faboya, O. M. (2020), Health care expenditure and child mortality in Nigeria, *International Journal of Health Care Quality Assurance*, 33(3), 261-275 Doi: 10.1108/IJHCQA-10-2019-0172.
- Akpesiri, E., & Eburajolo, C. (2022). Impact of monetary policy on the health sector in Nigeria: a special case of the infant mortality rate, *Ye-Numu. Journal of Economic and Development*, 5 (1), 38-60. <https://journals.fuotuo.ke.edu.ng/index.php/yjedi/article/view/150/150>
- Aonat, S. (n.d.). The role of technology in Nigeria's healthcare, *Slum and Rural Health Initiative Network*. Retrieved January 30, 2025, from <https://srhin.org/the-role-of-technology-in-nigerias-healthcare/>
- Attamah, N. (2018). Impact of government health sector expenditure on infant mortality in Nigeria (1981-2018), *International Digital Organization for Scientific Research* 3 (1), 194-207. <https://www.idosr.org/wp-content/uploads/2021/03/IDOSR-JBESS-31-194-207-2018-1.pdf>
- Awoyemi, B. O., Makanju, A. A., Mpapalika, J., Ekpeyo, R. S. (2023). A time series analysis of government expenditure and health outcomes in Nigeria, *Journal of Public Health in Africa* 2023; volume 14:1409. doi:10.4081/jphia.2023.1409
- Bala-gbogbo, E. (2025). Nigeria's economy grows at fastest in three years in fourth quarter. *Reuters*. <https://www.reuters.com/world/africa/nigerias-economy-grows-fastest-three-years-fourth-quarter-2025-02-25/>
- Boachie, M. K., Pölajeva, T., & Frimpong, A. O., (2020). Infant mortality in low-and middle-income countries: does government health spending matter? *Journal of Development Policy and Practice* 5(1), 54-73. <https://doi.org/10.1177/2455133320909916>
- Jiang, W., & Liu, X. (2022). *Infant mortality and inflation in China: Based on the Mixed Frequency VAR Analyses, Volume 10* <https://doi.org/10.3389/fpubh.2022.851714>

- Koginam, O. A., & Abraham, A. (2023), Economic growth and dimensions of mortality in Nigeria, *Fuoye Journal of Finance and Contemporary Issues* 5 (1). <https://fjfc.fuoye.edu.ng/index.php/fjfc/article/view/116>
- Macrotrends. (2024). Nigeria infant mortality rate 1950-2024. *Macrotrends*. <https://www.macrotrends.net/global-metrics/countries/NGA/nigeria/infant-mortality-rate>
- Mass life Healthcare Limited. (2025, February 7). *Rethinking Healthcare Funding in Nigeria: A Call for Self-Reliance*. Mass life Healthcare Limited | Quality and Affordable Healthcare Financing at Is Best! <https://masslife.com.ng/rethinking-healthcare-funding-in-nigeria-a-call-for-self-reliance/>
- Mishiwo, S., Adebajo, S., Banchani, E., & Adejumo, B. (2023). Infant mortality rate and economic growth in Nigeria: An empirical analysis. *International Journal of Innovative Finance and Economics Research* 11(3) 175-186. https://www.researchgate.net/publication/374389053_Infant_Mortality_Rate_And_Economic_Growth_In_Nigeria_An_Empirical_Analysis
- Mohammed, I. (2024). Nigeria's malnutrition crisis reveals the extent of its healthcare collapse. *The New Humanitarian* <https://www.thenewhumanitarian.org/news-feature/2024/10/28/nigeria-malnutrition-crisis-reveals-extent-healthcare-collapse>
- Naik, Y., Baker, P., Walker, I. Tillmann, T., Bash, K., Quantz, D., Brown, H. F., & Bamba, C. (2017). The macro-economic determinants of health and health inequalities, *Umbrella review protocol. Syst Rev* 6, 222. <https://doi.org/10.1186/s13643-017-0616-2>
- National Bureau of Statistics. (2019). *States nominal gross domestic product (2013 - 2017) - PHASE II*. https://www.nigerianstat.gov.ng/pdfuploads/State_Nominal_GDP_2013_-2017.cdr%28MAY_2019%29.pdf
- Raji, R. O. (2020). Infant and under five mortality rates in Nigeria: An impact analysis of macroeconomic conditions, *Risk and Financial Management* 2 (1), p1-p1. Doi:10.30560/rfm.v2n1p1
- Ritchie, H. (2021). *Definition: access to electricity*. Our World in Data. <https://ourworldindata.org/definition-electricity-access#article-citation>
- Solow, R. M. (1956). A contribution to the theory of economic growth, *The Quarterly Journal of Economics*, 70(1), 65–94. <https://doi.org/10.2307/1884513>
- Ubi, P., & Ndem, B. (2019). Poverty and health outcomes in Nigeria, *International Journal of Economics and Financial Issues*, 9(6), 132–141. <https://www.econjournals.com/index.php/ijefi/article/view/8704>

- Udemba, E. N., Emir, F., & Philip, L. D. (2023). Environmental implication of energy policies and private and public subsidies on infant mortality rate: a sustainable development study of India. *Environmental Science and Pollution Research* 30, 78680–78691. Doi: 10.1007/s11356-023-27981-4
- United Nations. (2015). *Transforming our world: The 2030 agenda for sustainable development*, United Nations General Assembly, A/RES/70/1. Retrieved from <https://sdgs.un.org/2030agenda>
- Uzor, K. J., Olarewaju, M. L., Asishana, T. K., Effiong, A., & Amaechi, A. U. (2024). The impact of inflation on medicine prices in Nigeria: a comparative analysis of public and private pharmacies. *Pan African Medical Journal*. 49:56. Doi: 10.11604/pamj.2024.49.56.45188
- World Bank. (2023). *Poverty and inequality*, The World Bank. <https://datatopics.worldbank.org/world-development-indicators/themes/poverty-and-inequality.html>
- World Bank. (2024). *Staying the course on reforms: Progress amidst challenges*, World Bank. <https://www.worldbank.org/en/news/press-release/2024/10/17/nigeria-staying-the-course-on-reforms-progress-amidst-challenges>
- World Bank. (2025). *Mortality Rate, infant per 1,000 live births.*, <https://databank.worldbank.org/indicator/SP.DYN.IMRT.IN>